Mathematical Foundations for Computing

Module 1
Module 1: Sets and Set Theory

Lesson 1.0 Introduction
- Course structure and navigation
  Reading • 15 min
- How to learn effectively on this course
  Reading • 15 min
- Introduction to the specialisation and this course
  Video • 4 min
- **Course Syllabus**
  Reading • 10 min

Lesson 1.1 Introduction to set theory
- The definition of a set
  Video • 8 min
- The definition of a set
  Reading • 10 min
- The definition of a set
  Practice Assignment • 30 min
- The listing method and rule of inclusion
  Video • 8 min
- The listing method and rule of inclusion
  Practice Assignment • 30 min
- The powerset of a set
  Video • 10 min
- The powerset of a set
  Practice Assignment • 15 min
- Set operations
  Video • 10 min
- Set operations
  Practice Assignment • 30 min

Lesson 1.2 Set representation and manipulation
- The representation of a set using Venn diagrams
  Video • 4 min
- The representation of a set using Venn diagrams
  Practice Assignment • 30 min
- Mastering Set Operations and Venn Diagram Reasoning
  Dialogue • 30 min
- De Morgan's laws
  Video • 8 min
- De Morgan's laws
  Practice Assignment • 30 min
- Laws of sets: Commutative, associative and distributives
  Video • 11 min
- Laws of sets: Commutative, associative and distributives
  Reading • 15 min
- Laws of sets: Commutative, associative and distributive
  Practice Assignment • 30 min
- Partition of a set
  Video • 3 min
- Partition of a set
  Practice Assignment • 30 min

Lesson 1.3 Summary and Assessment
- Module 1 summary
  Reading • 15 min
- Check your understanding: End of module 1
  Graded Assignment • 20 min

Module 2
Module 2: Number Systems and Bases

Module 3
Module 3: Functions and Their Properties

Module 4
Module 4: Relations

Lesson 4.1 Understanding the concept of relations
- Definition of a relation: relation versus function
  Video • 8 min
- Definition of a relation: relation versus function
  Practice Assignment • 20 min
- Matrix and graph representations of a relation
  Video • 10 min
- Matrix and graph representations of a relation
  Practice Assignment • 20 min
- The properties of a relation: reflexive, symmetric and anti-symmetric
  Video • 14 min
- The properties of a relation: reflexive, symmetric and anti-symmetric
  Practice Assignment • 20 min
- Relation properties: transitivity
  Video • 5 min
- Relation properties: transitivity
  Practice Assignment • 20 min

Lesson 4.2 Equivalence, and partial and total order relations
- Equivalence relations and equivalence classes
  Video • 6 min
- Equivalence relations and equivalence classes
  Practice Assignment • 25 min
- Partial and total order
  Video • 6 min
- Partial and total order
  Practice Assignment • 30 min
- Building and analysing different types of mathematical relations
  Dialogue • 15 min

Lesson 4.3 Summary and Assessments
- Topic summary
  Reading • 15 min
- Check your understanding: End of module 4
  Graded Assignment • 20 min

Course Wrap-Up
- Mathematical Foundations for Computing: Course Summary
  Reading • 10 min
- The Archive Room: A Mathematical Escape Mystery
  Role Play • 15 min
- Course summary
  Video • 1 min

Course Syllabus

0:06 / 5:57 2x

Course Description

Mathematics provides the formal structures and reasoning tools that underpin computer science. Mathematical Foundations for Computing introduces core topics essential for problem-solving, algorithm design, and theoretical computing. You will explore sets and set theory, number systems and bases, functions, and relations—building a toolkit to model data, describe systems, and reason about computational processes. Each module connects abstract mathematics to practical computing contexts, from binary representation and function mapping to relational models. By the end of the course, you will not only understand these concepts theoretically but also know how to apply them to programming, algorithms, and data structures. This course serves as the cornerstone of the Essential Mathematics for Computer Science specialisation, establishing the fundamental mathematical language and concepts necessary for success in advanced topics in logic, algorithms, and computational complexity.

Target Audience

This course is designed for computer science students, programming enthusiasts, and professionals seeking to strengthen their understanding of the mathematical principles underlying computing. It's ideal for:

- Computer science undergraduates needing a solid mathematical foundation
- Self-taught programmers looking to fill gaps in their theoretical knowledge
- Software professionals seeking to understand algorithms and data structures at a deeper level
- Anyone preparing for advanced study in computer science, artificial intelligence, or data science

No advanced mathematical background is required, though familiarity with basic algebra and arithmetic concepts will be helpful.

Content Differentiation

This course uniquely bridges theoretical mathematics with practical computing applications. Rather than presenting abstract mathematical concepts in isolation, each topic is directly connected to its application in computer science. The curriculum combines:

- Rigorous mathematical foundations with clear explanations of their relevance to computing
- Visual representations and interactive examples to build intuition for abstract concepts
- Progressive skill-building from fundamental definitions to complex applications
- Practice exercises that reinforce concepts through computational problem-solving

Through hands-on coding exercises and practical examples, you'll see how mathematical structures like sets, functions, and relations form the basis for data structures, algorithms, and computational models used in everyday programming.

Course Goals and Objectives

Upon successful completion of this course, you will be able to:

- Apply set theory concepts to model and solve computational problems
- Convert between different number bases and understand their significance in computing
- Analyze functions and their properties to evaluate computational processes
- Use relations to represent structured connections in computational systems
- Implement mathematical concepts directly in programming applications
- Apply formal mathematical reasoning to algorithm design and analysis
- Build a strong foundation for advanced topics in computer science mathematics

Course Outline

The course consists of four modules that progressively develop your understanding of mathematical foundations for computing:

Module 1: Sets and Set Theory Explores the fundamental concept of sets, set operations, Venn diagrams, and set properties essential for understanding data structures and logical operations.

Module 2: Number Systems and Bases Examines different number bases, conversion techniques, and operations particularly relevant to computer architecture and data representation.

Module 3: Functions and Their Properties Focuses on function definitions, properties, compositions, and special functions that model computational processes and transformations.

Module 4: Relations Investigates relations and their properties, equivalence relations, and order relations as frameworks for organizing and structuring data in computing.

Learning Objectives

Module 1: Sets and Set Theory

By the end of this module, you will:

- Define a set, its elements, and calculate set cardinality
- Explain set operations including union, intersection, difference, and symmetric difference
- Apply De Morgan's laws using membership tables
- Create and interpret Venn diagrams to visualize set relationships
- Understand universal sets, complements, and powersets
- Use set partitioning in computational contexts

Module 2: Number Systems and Bases

By the end of this module, you will:

- Explain number bases and their importance in computing
- Convert integers and non-integers between decimal, binary, and other bases
- Perform binary arithmetic operations (addition, subtraction, multiplication)
- Distinguish between rational and irrational numbers in different representations
- Apply number system concepts to computer architecture problems

Module 3: Functions and Their Properties

By the end of this module, you will:

- Define functions and distinguish them from other relations
- Identify and prove injective, surjective, and bijective functions
- Compose functions and understand their applications
- Analyze special functions including logarithmic, floor, and ceiling functions
- Plot functions and interpret their graphical representations
- Apply function concepts to algorithm analysis

Module 4: Relations

By the end of this module, you will:

- Define relations and compare them to functions
- Represent relations using matrices and digraphs
- Analyze relation properties including reflexivity, symmetry, anti-symmetry, and transitivity
- Identify and work with equivalence relations and equivalence classes
- Apply partial and total order relations to computational problems
- Use relations to model dependencies in computational systems

Go to next item ✓ Completed

Like Dislike Report an issue

Go to next item →